The problem: 6 months of HR exports, no trusted answers

- Indus Weave Textiles: 1,266 employees, 4 plants, 12 HCM files (Apr-Sep 2026) under 21 contradictory business rules.
- The export is deliberately messy: mixed date formats, 50 duplicate employees, 37 broken history rows, a biometric clock running 30 minutes slow.
- The CFO needs answers it can defend: headcount trend, attrition, overtime authenticity, and who is being paid for what.
- Our job: one pipeline that turns these files into evidence - and proves the evidence is clean.

The pipeline: one command, six gated stages

python run_pipeline.py

raw -> staging -> attendance -> warehouse -> 10 queries -> 58 DQ checks (~2 min)

- Raw is immutable: files load verbatim (TEXT + hash), then Rules 1-21 repair them in code - data/ is never edited.
- Attendance engine: sessionised punches, device-drift corrected, night shifts attributed to their start date; OT rounded down to half-hours.
- Star schema: 4 slowly-changing dimensions + 4 fact tables, all foreign keys verified (0 orphans), ERD included.
- Quality gate: 58 automated checks (reconciliation, keys, SCD2, domains) written to dq_results - currently 58 PASS / 0 FAIL, proven to FAIL when seeded bad data.

Gates R -> S/A -> W -> Q -> D all PASS | 32/32 tests green | SQLite, no server

Finding 1: Stitching pays overtime it cannot see

+2,758 h

paid vs punch-computed OT in Stitching = +56% (approx. PKR 1.36 m unsupported)

- Only department over the 15% threshold: paid 7,659 h / PKR 3,771,337 vs 4,901 h justified by punches.
- The gap is zero in April and grows every month to +130% in September - an escalation, not rounding.
- Approved hours track PAID hours (7,656), not punch hours: approvals ratify the payment instead of the work.
- Action: block OT posting unless approved hours sit inside punch-derived OT windows; review May-Sep D20 approvals.

Evidence: analytics/q06_dept_paid_over_computed_15pct.sql | 1 row out of 25 departments

Finding 2: Pay for people who are never at work

PKR 2,879,197

7 employees, 37 employee-months paid with ZERO present days

- Q7 finds 58 employee-months (PKR 4.83 m) paid with zero worked hours; 37 of them (PKR 2.88 m) are people still on the payroll, not leavers.
- Six of the seven were never Present in any of the six months while salary kept flowing - e.g. E10399: 6 months, PKR 541,125, 0 present days.
- Zero Present + zero Absent rows means these employees sit outside the workforce spine entirely: invisible to HR, visible to payroll.
- Action: attendance-to-payroll release gate - no month pays while worked_hours = 0 exists uncertified.

Evidence: analytics/q07_paid_zero_attendance.sql | 58 rows | overlaps Finding 3 by 21 months

Finding 3: Payroll keeps paying people who left

PKR 1,948,000

5 ex-employees, 21 FULL months paid after their separation date

- Every payroll month that starts after separation_date still pays earnings - final settlement months excluded, so this is pure overrun.
- Worst case: E10471 left 23 Apr, collected PKR 132,000/month for May-Sep (131 days, PKR 660,000) with zero attendance.
- Four of the five left in mid-May yet were paid in the June run too - the offboarding trigger failed twice, not once.
- Action: automatic payroll block where separation_date < month start (the query IS the predicate) + recoveries through the exit checklist.

Evidence: analytics/q08_pay_after_separation.sql | 21 rows | months starting after last day worked

The ask

- Approve three automated controls, all already written as queries: OT-must-match-punches, zero-attendance pay block, post-separation payroll block.
- Total exposure identified in ONE six-month window: approx. PKR 6.2 m (1.36 + 2.88 + 1.95) - findings overlap only where noted.
- Run python run_pipeline.py monthly: every number regenerates, every DQ check re-runs, dq_results is the audit trail.
- Deliverables: this deck, cfo_findings.md (3 findings with evidence SQL), README, ERD, 10 questions, 58 DQ checks, 32 tests.